

Products and coproducts

The slightly more advanced universal properties of products and coproducts, together with examples in various of the categories we've seen.

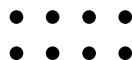
Products and coproducts are our next universal property, slightly more complicated than terminal and initial objects. I have mentioned that every universal property is an initial object *somewhere*, and indeed products and coproducts in a category $\mathcal{C}$ are “just” terminal and initial objects in a “souped-up” category based on $\mathcal{C}$. However, it is often more helpful to translate a universal property back into $\mathcal{C}$ rather than thinking about that other souped-up category all the time. This is like when we thought about dual concepts by placing a concept in $\mathcal{C}^{\text{op}}$, but we translated the concept back into $\mathcal{C}$. I think it's like when you make a paper snowflake by folding up a piece of paper and cutting out a pattern — the fun part is opening out the paper again.

We will start with products, and then see the dual concept, coproducts.

18.1 The idea behind categorical products

In basic math the product of two numbers a and b is a number $a \times b$ that has a certain relationship with the original two numbers. When introducing this to children we might start by investigating $2 + 2$, and then $2 + 2 + 2$, and then $2 + 2 + 2 + 2$, and we might say that the first one is “two times two” and the next is “three times two” and the next is “four times two” and so on. Then there's the tricky question of why 4×2 is the same as 2×4 , when the first is $2 + 2 + 2 + 2$ and the other is $4 + 4$.

We might then get them to play with some small objects and line up four rows of two in a grid. Then if you rotate the grid you see two rows of four instead.



So $a \times b$ can be thought of as a lots of b , or b lots of a , or the number of things in a grid of a rows and b columns, or the number of things in a grid of b rows and a columns. Some people argue about which view of multiplication is “best”

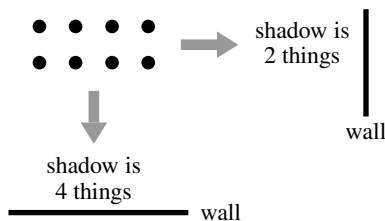
but I think the best thing (if we must make it a competition) is to understand all points of view and their relationships with each other.

Expressing multiplication of numbers by repeated addition does not generalize very well, either to more complicated types of number (like fractions or irrational numbers) or to more complicated types of object such as shapes, and this is where category theory comes in. We're going to look for a general type of multiplication that we can attempt in any category. It won't necessarily be *possible* in every category, just like the fact that some categories have terminal and initial objects and some don't. Some categories have products and some don't. It's pretty handy when they do.

For categorical products we will start with two objects a and b in our category, and produce a third object $a \times b$ which deserves to be called a product of a and b by virtue of certain relationships it has with a and b . The question is what those relationships are.

If we think about our 2 by 4 grid of objects above, what is the relationship between this grid and the numbers 2 and 4? We might say there are 2 rows and 4 columns, but that sort of presupposes an understanding of what rows and columns are. We might say its height is 2 and its width is 4, but that also presupposes concepts of height and width. Here is something we could say that's (perhaps) more fundamental.

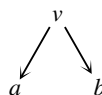
Imagine projecting shadows of our objects onto a wall. If we project in one direction (and perfectly line everything up) then we will see only 2 things, and if we project in the other direction we'll see 4.



This is pretty much the idea of how we formalize products in category theory.

18.2 Formal definition

Given objects a and b we are going to think about configurations of arrows as shown on the right; the two arrows are like the two arrows in the shadow diagram above.

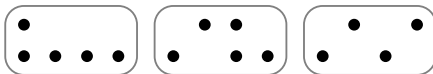


But we don't want any old thing that will do this: we want a universal one. This means we want it to be somehow "canonical", or inherent, or essential.

Things To Think About

T 18.1 Can you think of some irregular configurations of objects whose shadows as above still produce 2 things in one direction and 4 in the other?

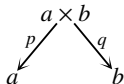
Here are some configurations that produce 2 and 4 object shadows in sneaky, non-canonical ways.[†]



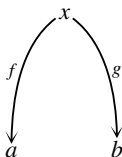
The full definition of product in a category demands that we have a diagram with two “projection” morphisms to a and b , and moreover that the diagram is universal among all such diagrams, in the following sense.

Definition 18.1 Given objects a and b in a category, a *product* for them is:

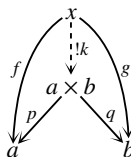
an object $a \times b$
equipped with
morphisms p and
 q as shown below



such that given any
diagram of this form



there is a unique morphism
 $x \xrightarrow{k} a \times b$ making the
following diagram commute.[‡]



This is a standard format for a universal property. The second diagram is abstractly the same shape as the first, I’ve just drawn it differently physically to emphasize how they fit together in the third diagram; the second is often not drawn as we might just say “given any such diagrams...”. The third diagram says that the first is universal among all diagrams of the same shape, so once we understand what that means we might not draw that one either.

The morphism k in the third diagram is a type of comparison between the universal situation (the first diagram) and the non-universal one (the second diagram). Note that asking for the diagram to commute means that the triangle on each side commutes. Written out in algebra this means $pk = f$ and $qk = g$ but I think it’s really better to think of it geometrically in the diagram.

Finally note that the uniqueness of the morphism k needs to be taken together with the “such that” clause afterwards: it might not be the only morphism $x \rightarrow a \times b$, but it has to be the only morphism that makes both of those side triangles commute.

[†] This reminds me of the image on the cover of Douglas Hofstadter’s spectacular book *Gödel, Escher, Bach: An Eternal Golden Braid* featuring a carved three-dimensional object which miraculously manages to project the shadows G, E, B, and also E, G, B. If you don’t know it, I urge you to look it up, partly just because it’s an amazing book and partly because the image on the cover is impossible to describe in words.

[‡] The ! symbol signifies “unique”.

Terminology

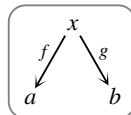
This sort of situation is going to come up repeatedly so it's useful to have some terminology.

- The morphisms p and q are called *projections*.
- The morphism k is called a *factorization*. We are factorizing f and g .
- There is a general type of diagram called a *cone* which consists of a *vertex* and the morphisms from that vertex to everything else. In this case the vertex is $a \times b$ and “everything else” is just the objects a and b .
- The product is the *universal cone*, universal among all cones of this shape.
- We say the universal property *induces* the factorization k .

18.3 Products as terminal objects

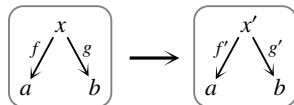
Products are in fact a special type of terminal object. It's just that a product in $\mathcal{C}$ isn't a terminal object in $\mathcal{C}$, it's a terminal object in some more complicated category based on $\mathcal{C}$, which I like to call a “souped-up” category. We already saw this principle for terminal objects themselves: terminal objects in $\mathcal{C}$ are initial objects in $\mathcal{C}^{\text{op}}$. We are generally going to see that we can make a more complicated category based on $\mathcal{C}$, look for a simple structure in there, and then translate it back to a structure directly in $\mathcal{C}$.

For products, we need a category whose objects are whole diagrams of the shape in question. That is, given objects a and b whose product we want to define, we make a new category whose objects are diagrams of the form shown here.

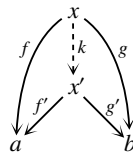


Note that a and b stay the same, but we have a choice of x , f and g .

Now we need to define morphisms, for example between the objects shown here. Remember that the diagram in each box counts as one object in this “souped-up” category.



A morphism is going to look like the diagram in the definition of product. It consists of a morphism $x \xrightarrow{k} x'$ making the diagram on the right commute. We can think of it as a “structure-respecting map” where “respect” means “make everything commute”.

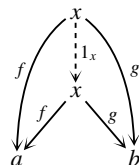


Identities, composition and axioms are “inherited” from $\mathcal{C}$, which makes sense because morphisms in this category are just some particular morphisms in $\mathcal{C}$.

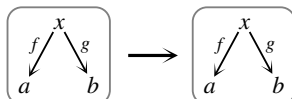
Things To Think About

T 18.2 What would we need to do to make this rigorous?

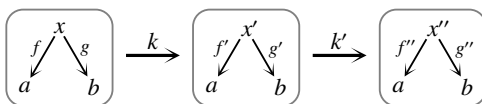
This is an example of when it's almost more important to understand what the question is than what the answer is. For identities we just need to observe that the diagram on the right commutes.



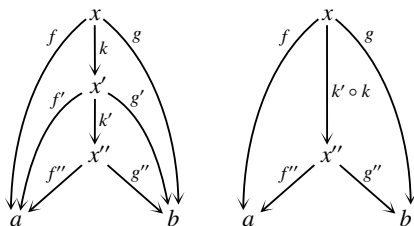
This shows that the morphism 1_x gives a valid morphism from this object to itself:



For composition we need to consider the type of scenario shown on the right.

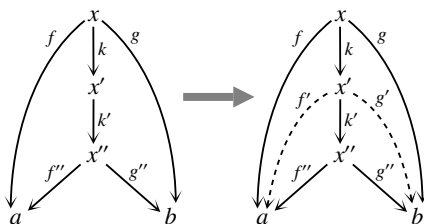


This means we have the situation shown in the first diagram on the right. What we need to know is that the composite $k' \circ k$ is a valid morphism from the beginning to the end, which means we need to show that the second diagram commutes.



This is true because if you stick together commutative diagrams you get commutative diagrams. That's rather vague, but the thought process typically goes as follows.

We want to show that the first diagram on the right commutes, so we “fill it in” with the dotted arrows as shown in the second diagram. (This is where it would really help to be a video.) Then because all the small areas commute, we know that the larger diagram commutes.



If we wrote it out in algebra it would look like this:

$$\begin{aligned}
 f'' \circ k' \circ k &= f' \circ k && \text{as the bottom left triangle commutes} \\
 &= f && \text{as the top left triangle commutes} \\
 g'' \circ k' \circ k &= g' \circ k && \text{as the bottom right triangle commutes} \\
 &= g && \text{as the top right triangle commutes}
 \end{aligned}$$

I hope you can see that the method using diagrams is much more illuminating while containing all the information, plus a little more.

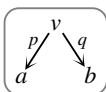
Proving things in category theory involves a lot of trying to show that some large diagram commutes by “filling it in” in this way. This is what is known as “making a diagram commute”.[†] Personally I find it really fun and satisfying in a way that strings of symbols are not (to me). I appreciate being able to invoke geometric intuition. Moreover, a string of symbols omits the source and target information for the morphisms and so loses a possibility for type-checking. For example, in the above argument it is possible to write the symbols $k \circ k'$ by mistake, although those morphisms are not composable that way round. Whereas if you draw them out as arrows, then as long as you get their sources and targets right you can’t physically draw the composite the wrong way round.

Things To Think About

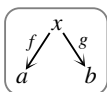
T 18.3 Can you see how terminal objects in our “souped-up” category correspond to products in $\mathcal{C}$?

A terminal object in this category is:

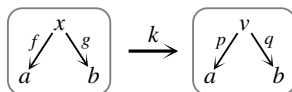
an object



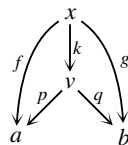
such that for any object



there is a unique morphism



But such a morphism k in the souped-up category is a morphism $x \xrightarrow{k} v$ in $\mathcal{C}$ making the diagram on the right commute, which is exactly the unique factorization we need for the definition of product.



Note that k is both a morphism in $\mathcal{C}$ and, because it makes the appropriate diagram commute, a morphism in the souped-up category.

Things To Think About

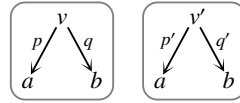
T 18.4 As the product is a terminal object, we get some uniqueness from the uniqueness of terminal objects. Can you unravel that and say what it is in $\mathcal{C}$?

We know that terminal objects are unique up to unique isomorphism, and products are a terminal object *somewhere*, so products must also be unique up to

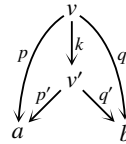
[†] Some more pedantic people object to this term on the grounds that we’re not making it commute, we’re showing that it commutes.

unique isomorphism. However we have to be careful what that means: it's only going to be unique in the *somewhere* category, not in $\mathcal{C}$.

That is, in our souped-up category of diagrams, if the two objects shown here are both terminal then there must be a unique isomorphism between them *in that category*.



This means there is a unique isomorphism $v \xrightarrow{k} v'$ making the diagram on the right commute. Importantly it is not just a unique isomorphism between the vertices v and v' . There might be many isomorphisms $v \rightarrow v'$, but there will only be one that makes the diagram commute.



This is the idea of uniqueness for all universal properties. A product in a category is not just an object $a \times b$: it's an object together with the projection morphisms, so uniqueness has to be up to unique isomorphism respecting those.[†] Finally note that we write the product of a and b as $a \times b$ although many different (isomorphic) objects could be regarded as $a \times b$. We'll see this now with examples in **Set**.

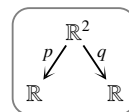
18.4 Products in **Set**

You might well have seen some products in **Set** in your life, even if you didn't realize it. Every time we draw a graph with an x -axis and a y -axis we're using a product. If each axis is the real numbers $\mathbb{R}$ then the 2-dimensional plane is a product $\mathbb{R} \times \mathbb{R}$, also written $\mathbb{R}^2$. Using axes in this way is named “cartesian” after Descartes, and this type of product is generally called a “cartesian product”.

Things To Think About

T 18.5 Can you see how $\mathbb{R}^2$ is a categorical product $\mathbb{R} \times \mathbb{R}$ in the category **Set**? See if you can work out what the projections are, and how the universal property works. Remember that an object of the cartesian plane is a point given by a pair of coordinates (x, y) .

To show that $\mathbb{R}^2$ is a categorical product first we need to exhibit the projections, that is, give the whole diagram on the right. The idea here is that one of the copies of $\mathbb{R}$ in the “feet” is the x -axis and the other copy is the y -axis.

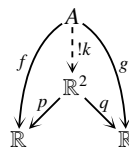


[†] This didn't arise with terminal and initial objects because there weren't any projection morphisms, so “respecting those” was vacuously satisfied.

The projections p and q then project onto the x - and y -coordinate respectively, as specified here.

$$\begin{array}{ccccc} \mathbb{R} & \xleftarrow{p} & \mathbb{R}^2 & \xrightarrow{q} & \mathbb{R} \\ x & \longleftarrow & (x, y) & \longrightarrow & y \end{array}$$

The universal property then says: given any diagram as in the curved part on the right, we have a unique factorization k as shown. (Again, the $!$ symbol signifies “unique”.)



This says that for any set A , a pair of functions f and g as shown amounts to a function producing points in the cartesian plane, as we can interpret f and g as producing an x - and y -coordinate respectively. This is an important interpretation of the universal property. Whenever we use the phrase “given any...there exists a unique...” we’re saying, among other things, that there’s a bijective[†] correspondence between the first type of thing and the second type of thing. For products it means we have a bijective correspondence between:

- pairs of individual morphisms to the “feet”, and
- a single morphism to the vertex.

The fact that the bijection is produced via composition with a factorization makes it even better, and we’ll come back to the meaning of that. I think understanding this “meaning” of the universal property is at least as important as grasping the technical procedure.

Things To Think About

T 18.6 See if you can formally construct the factorization, using the ideas above, and prove that it is unique. As usual, to show uniqueness we can suppose there is another one, say h , and demonstrate that we must have $h = k$.

For the above example with $\mathbb{R}^2$ we construct the unique factorization k from f and g as follows: given any object $a \in A$ we need to specify the point $k(a) \in \mathbb{R}^2$. That point has its x -coordinate produced by f and its y -coordinate produced by g . Formally it’s this: $k(a) = (f(a), g(a))$.

Now we can *sense* it’s the unique one making the diagram commute because we had no choice about how to do it. We knew that k had to make the two sides of the diagram commute; the left-hand triangle says doing k and then taking the x -coordinate has to give the same result as doing f , and similarly for the right-hand triangle doing k and taking the y -coordinate has to be the same as doing g . That is more or less the whole proof.

[†] Reminder: a bijective correspondence is one that pairs things up perfectly.

Proposition 18.2 *The function k defined above is the unique morphism making the diagram defining the universal property commute.*

Proof Consider any morphism k making the diagram commute. By definition of p and q , we know $k(a) = (pk(a), qk(a))$.

- We know that the left-hand triangle commutes so $pk(a) = f(a)$.
- Similarly from the right-hand triangle we get $qk(a) = g(a)$.
- Thus $k(a) = (f(a), g(a))$. □

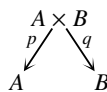
This always reminds me of an Etch-a-Sketch: you have one dial controlling your x -coordinate, and another dial controlling your y -coordinate, and then if you're very well coordinated you can draw pictures of anything at all in the 2-D drawing space, by controlling each coordinate separately. It is just one example of general cartesian products in **Set**. Given any two sets A and B we can make a set $A \times B$ that is something like making “ A -coordinates and B -coordinates”.

Things To Think About

T 18.7 See if you can come up with a definition of what this product set is.

Definition 18.3 Let A and B be sets. Then the *cartesian product* $A \times B$ is the set defined by $A \times B = \{ (a, b) \mid a \in A, b \in B \}$.

The elements (a, b) are called *ordered pairs*, and there are functions p and q as shown on the right called *projections*, sending an element (a, b) to a and b respectively.



We saw ordered pairs in Section 15.2 when we looked at the example of Lisa Davis in New York. We were thinking about pairs “name and date of birth”, so we could say the set A was the set of names, the set B was the set of dates of birth, and the function in question was from the set of people to the cartesian product $A \times B$.

Things To Think About

T 18.8 Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$.

1. Write out all the elements of $A \times B$. Can you lay them out on the page in a way that shows how it's related to products of numbers?
2. Can you think of some places in normal life where this sort of thing arises?

Here are the elements of $A \times B$. I have laid them out in a grid to relate it back to the grids I drew at the start of the chapter for multiplication of numbers. I've also put in “axes”, to make it look even more like coordinates.

b	$(1, b)$	$(2, b)$	$(3, b)$
a	$(1, a)$	$(2, a)$	$(3, a)$
	1	2	3

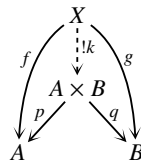
This is now very similar to how squares on a chessboard are typically labeled for reference, with the numbers 1–8 in one direction and the letters A–H in the other. It’s also how maps used to have grids on them back when we used to look up a street in an index — it would say something like “p.27, 4A” and we’d then have to look in that square to try and find the street we wanted.

It’s also why I like drawing a grid of a hundred numbers from 0 to 99 (as in Chapter 3) rather than 1 to 100, because that way we basically get to see the two digits of the numbers as coordinates. Whereas if you start at 1 then the rows won’t have a consistent first coordinate; for example the row starting 11 will end with 20, changing the first digit.

Things To Think About

T 18.9 Prove that the cartesian product together with its projections is a categorical product of A and B . This is exactly like the proof we did for $\mathbb{R}^2$ but just with general sets A and B instead. Can you also prove that $B \times A$ is a categorical product of A and B ? What is the unique isomorphism with $A \times B$?

Proof Consider morphisms f and g as in the curved part of the diagram on the right; we need to show that there is a unique factorization k as shown. Now, in order for the diagram to commute we must have $k(x) = (f(x), g(x))$ for any $x \in X$, and this definition makes k a unique factorization.



This shows that $A \times B$ is a categorical product as claimed. $\square$

Note that we have streamlined this argument somewhat, basically saying “if there were a function making these triangles commute, it would have to behave like this, and oh look, there is a function that does that” which means, again, we did the uniqueness part in advance. This will be even more noticeable when we do this for structure-preserving maps like group homomorphisms, because we will say “if there is a group homomorphism here it has to do this, and now we check that that is in fact a group homomorphism”.

Note also that we have given up drawing the diagram in stages. In the absence of books with diagrams that grow in front of our eyes (otherwise known as a video), we often draw a single diagram and read it “dynamically”. The dotted arrow is understood to come later, induced by the rest of the structure.

Now let’s think about the cartesian product $B \times A$. This is a set of ordered pairs but ordered the other way round: $B \times A = \{ (b, a) \mid a \in A, b \in B \}$.

There is then an isomorphism of sets switching between those two expressions, as shown here.

$$\begin{aligned} A \times B &\longrightarrow B \times A \\ (a, b) &\longmapsto (b, a) \end{aligned}$$

This is the unique isomorphism between those two sets *as products*; there are many other isomorphisms between them as sets. We will now specifically examine uniqueness for products of sets.

18.5 Uniqueness of products in **Set**

We previously saw that any object isomorphic to an initial object is also initial (and likewise for terminal objects). We have also seen that uniqueness of products is more subtle: the unique isomorphism between products is only unique *such that* the diagram involving the projections commutes. We will now investigate this for products in **Set**, to see it working in practice.

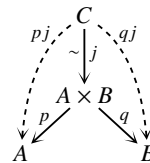
Things To Think About

T 18.10 Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$ again as in the earlier example, and let $C = \{1, 2, 3, 4, 5, 6\}$.

1. Show that C is isomorphic to the cartesian product $A \times B$. How many isomorphisms are there?
2. Show that each isomorphism uniquely equips C with the structure of a categorical product of A and B . This means that it is equipped with projection maps and has a universal property.
3. What is the content of saying that $A \times B \cong C$ non-uniquely, but that when C has the structure of a product it is uniquely isomorphic to $A \times B$?

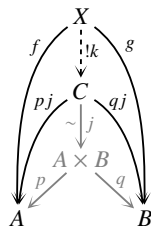
Proof The cartesian product $A \times B$ has 6 elements, and so does C , so we know they are isomorphic because finite sets are isomorphic if and only if they have the same number of elements. There are many isomorphisms however. We did some counting of isomorphisms in Section 14.5 and we can do something similar here to find that the number of possible bijections is $6 \times 5 \times 4 \times 3 \times 2 \times 1$ which is written as $6!$ and known as “6 factorial”.[†]

Now, given an isomorphism $C \xrightarrow{j} A \times B$ we can compose with p and q as shown to get *putative* projections for C . (“Putative” means that they are candidates for being the projections but we can’t say they definitely work until we check some more stuff; however usually when we say this it means that it is going to turn out to be right.)



[†] If we construct a bijection $C \longrightarrow A \times B$ then we could first pick where 1 goes, and we have 6 possibilities, and then we pick where 2 goes, for which there are 5 remaining possibilities, and so on. You may have seen this if you have ever studied how to count permutations of n elements; a permutation is precisely a bijection from a set to itself.

We now need to consider f and g as in the diagram on the right, and show that there is a unique factorization k as shown. In this diagram the parts in gray are to remind us how pj and qj are constructed, and the $\sim$ sign is to remind us that j is an isomorphism. All the arrows are functions.

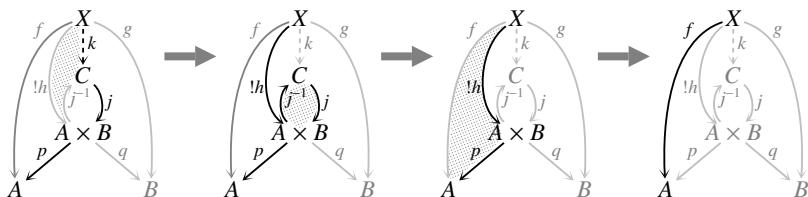


The thing to think now is: what ways are there to get from X to C ? We can get from X to $A \times B$ from the latter's universal property, and then we can go “backwards” up j as it is an isomorphism. We now make that formal.

The universal property of $A \times B$ induces a unique factorization $X \xrightarrow{!h} A \times B$ making a certain diagram commute, and we can then use the inverse j^{-1} of j to make a morphism $X \xrightarrow{!h} A \times B \xrightarrow{j^{-1}} C$.

We now have to check some things, but if you’re thinking categorically you will already *feel* that this must be correct. Eventually you wouldn’t need to see a proof like this beyond this point. See remarks on “abstract nonsense” after the proof.

We now have to check two things: that this really is a factorization (i.e. the relevant diagram commutes) and that it is the unique one. To show that the diagram commutes consists of a “diagram chase” where we follow arrows around a diagram progressively. It’s quite hard to show this in a static diagram on a page, but I will try. Again this is really all part of one diagram, but we’re reading it “dynamically”, working our way through it as shown. At each stage we’re really just considering the black arrows and not thinking about the gray ones at the moment. Then we notice that the shaded region commutes so we can move across it, which takes us to the black arrows in the next diagram.



Written out in algebra, it would look like this:

$$\begin{aligned}
 pj k &= pj j^{-1} h && \text{by definition of } k \\
 &= p h && \text{by definition of } j^{-1} \\
 &= f && \text{by definition of } h
 \end{aligned}$$

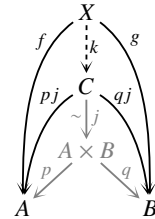
Each of the equals signs corresponds to one move across a large gray arrow to the next diagram. The explanation in words to the right corresponds to the gray shaded area in each diagram, which is telling us how we can move to the next diagram and thus to the next line of the algebra. I hope you can see how the algebra corresponds to the dynamic reading of the diagram, and that you can become comfortable with the diagrams, and then become comfortable with a single diagram read dynamically. This is a key technique in category theory.

This shows that k is indeed a factorization; we still need to show it is the unique one.

The idea here is that k is inextricably linked to h by construction, and h is unique so k “must be” unique too. To make this formal we can assume there’s a different one and show that this would produce a different version of h too.

The key is the diagram on the right, showing how we constructed the projections for C and the factorization k .

If you read the diagram dynamically you can see that k is a factorization for the product C precisely if jk is a factorization for the product $A \times B$. Thus if k' is also a factorization for the product C then jk' is also a factorization for the product $A \times B$.



But we already know that the factorization for $A \times B$ is unique, thus we must have $jk = jk'$, and then since j is an isomorphism it follows[†] that $k = k'$. $\square$

Note on “abstract nonsense”

Many category theory proofs proceed in this way, especially where universal properties are involved: you need to exhibit a piece of structure satisfying some properties, and once you’ve found a putative structure that type-checks, you *know* it’s going to satisfy the properties because of how you produced the structure. In the above proof to produce the factorization we used exactly the information available: the universal property of $A \times B$, and the isomorphism with C . There was no other way to get any sort of morphism $X \rightarrow C$; it was almost just type-checking that led us here. This is quite typical of category theory proofs and can sometimes lead people to think that they have no content, giving rise to the description “abstract nonsense”. This probably started as an

[†] If we wrote this out fully in algebra it would look something like this. Suppose k' is also a factorization, that is $(pj)k' = f$ and $(qj)k' = g$. But this means $p(jk') = f$ and $q(jk') = g$. But there is a unique morphism h such that $ph = f$ and $qh = g$, and so $jk' = h$. But we already knew $jk = h$, so $jk' = jk$, and since j is an isomorphism we can apply j^{-1} on the left of both sides of the equation to get $k' = k$. [I find the dynamic diagram much more enlightening.]

insult to category theory and then was reclaimed by some category theorists as an affectionate way of describing this process. I don't like using it, even in affectionate jest, because I don't like the word "nonsense". I think this is abstract profundity. The framework beautifully slots everything into place for us.

Finally I'll stress that there may be many isomorphisms $C \xrightarrow{\sim} A \times B$ as sets (as we saw with the small example producing 6! isomorphisms), but only one isomorphism as products. This means that once C has been equipped with the structure of a product, there is only one isomorphism with $C \xrightarrow{\sim} A \times B$ respecting that structure, that is, commuting with the projections.

There is something perhaps a little hazy in the notation of products in **Set**, as both categorical products and cartesian products are likely to be notated $A \times B$. However, cartesian products are just one particular construction of categorical products in **Set**, a particularly common one. One consequence is we end up having to be rather careful about things like associativity.

Aside on associativity

I will just mention this briefly: cartesian product is not strictly associative, for a slightly pedantic reason. For the two different bracketings we get:

$$\begin{aligned} (A \times B) \times C &= \{ ((a, b), c) \mid a \in A, b \in B, c \in C \} \\ A \times (B \times C) &= \{ (a, (b, c)) \mid a \in A, b \in B, c \in C \} \end{aligned}$$

Both ways round involve the same information, that is, ordered triples of elements, but written in a slightly different way. If you feel that the difference between these two things is just an annoyance then I agree, and it shows that we're working at the wrong level here. We are trying to ask about this equation: $(A \times B) \times C = A \times (B \times C)$. But this is an *equality* between objects in a category: we should be asking about isomorphisms. The two sets above are certainly isomorphic, in a particularly contentless way, but we need one more dimension to express that rigorously. It is the question of "weak associativity".

Finally note that for categorical products it's not just asking for the wrong level of sameness: it doesn't even make logical sense to pose the question, because the products $(A \times B) \times C$ and $A \times (B \times C)$ are only *defined* up to isomorphism. If something is defined up to isomorphism you can't ask if it equals something else — or, if you do, you have to be rather careful what you mean. In the case of categorical products the only thing we could mean is that each side of the "equation" canonically has the structure of the product on the other side. And to complicate things further, there is another three-fold product that is not derived from the binary products. For cartesian products it is this:

$$A \times B \times C = \{ (a, b, c) \mid a \in A, b \in B, c \in C \}.$$

Things To Think About

T 18.11 Try investigating the categorical three-fold product that corresponds to this. It will have three projections instead of just two.

This all becomes much more crucial when we move into higher dimensions. The more dimensions we have, the more crucially subtle it all becomes.

Terminology

It is not always possible to find products of objects in a category — there might be no diagrams with the desired universal property. If a category has enough structure that we can take products of *any* two objects in the category, then we say that the category *has binary products*. If it has binary products then by induction we can build up to n -fold products for any n , so we say that the category *has finite products*.

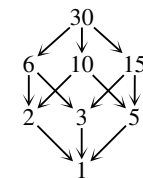
So far we have looked in quite some detail at products in **Set**. We will now take a look at products in some of the other categories we've seen.

18.6 Products inside posets

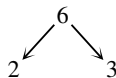
Recall that a poset is a category with at most one morphism between any two objects, and that this means all diagrams commute. So in a poset the definition of product simplifies: we don't have to check any commutativity or uniqueness for the factorization.

Things To Think About

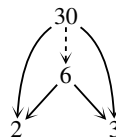
T 18.12 Recall the poset of factors of 30 on the right. Show that 6 is the categorical product of 2 and 3. This seems “obvious”, but don't get too complacent: what is the categorical product of 6 and 10 in this category? Can you generalize this to posets of factors of a general n ?



To show that 6 is a categorical product we need to exhibit projections and a universal property. We can see from the diagram that it has putative projections as shown here.



For the universal property we need to consider any number in the diagram that has morphisms to 2 and 3. The only number that does (other than 6) is 30, and we duly do have a factorization as shown here.



It might seem coherent that the categorical product of 2 and 3 is the “ordinary” product, but let’s now look at the categorical product of 6 and 10. There is only one candidate for product now because there is only one object with projections to 6 and 10, and that is 30; as it is the only one it must be universal.

The “souped-up category” in which we’re looking for a terminal object has only one object and only one (identity) morphism, so that object is definitely terminal.

There are now (at least) two ways we could proceed, to see how to understand and generalize this. They sort of correspond to experiment and theory: we could experiment with other posets of factors to see if we can see a pattern, or we could go through the definition of product in this particular context to understand what it produces.

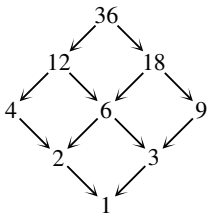
Things To Think About

T 18.13 Here are some suggestions for either of those thought processes.

1. If you want to explore other numbers, I suggest the diagram for factors of 36. What is the categorical product of 4 and 6 in there? What about 6 and 9? 4 and 9? 2 and 4?
2. If you want to go ahead and explore by theory, remember that a morphism $a \rightarrow b$ in this category (the way I’ve drawn it here) is the assertion that a is a multiple of b . Write out the definition of product using that definition in place of each morphism, and see what you get.

Here is the diagram for the poset of factors of 36. Remember we can work this out from the prime factorization $36 = 2 \times 2 \times 3 \times 3$ which tells us we have

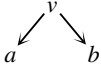
- two dimensions, one for 2 and one for 3, and
- paths of length two in each direction, as each prime factor appears twice in the factorization.



The categorical products can more or less be seen as the lowest point in the diagram that has a morphism to each object in question. So the pairs of numbers I asked about have categorical products as shown in this table.

		categorical product
4	6	12
6	9	18
4	9	36
2	4	4

If we now work through the definition of product in this context, we see that for the product of a and b we need:

1. A diagram of the form shown here, which in this category just means that v is a multiple of both a and b . 
2. Given any such diagram with vertex x , we need a morphism $x \rightarrow v$. Here is a “translation” of what that means in this particular category.

Given any such diagram with vertex x	we need a morphism $x \rightarrow v$
given x that is a multiple of both a and b	x is a multiple of v

These two thoughts together, translated, give exactly the definition of lowest common multiple. You might think that the lowest common multiple is usually defined to be the lowest among all common multiples, but in fact it follows that any other multiple is not only larger but is also a multiple of the lowest one, and this is the definition that is often used in more abstract mathematics.

In a general poset we might generically call the ordering $\geq$ and in that case the definition of categorical product gives us this:

1. an object v such that $v \geq a$ and $v \geq b$, and
2. for any object x such that $x \geq a$ and $x \geq b$, we must have $x \geq v$.

This is exactly the definition of *least upper bound* or *supremum*. The first point says that v is an upper bound for a and b , and the second point says that if x is any upper bound, it is greater than or equal to v . Least upper bounds are important in analysis and careful study of the real numbers, as the existence of least upper bounds is a way of distinguishing the reals from the rationals.[†]

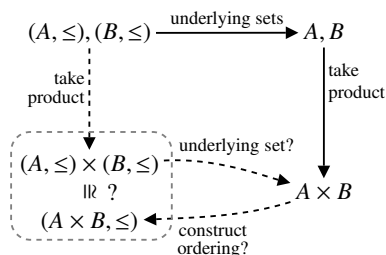
18.7 The category of posets

We will now zoom out, and instead of looking inside a poset expressed as a category, we will look at the category **Pst** of posets and order-preserving maps.

When looking for a universal property for sets-with-structure, a typical approach is to start with the universal construction on the underlying sets, and then see if there’s a natural way to extend the structure on them. We are making use of the system of interactions below. (The squiggly sign is an isomorphism

[†] A set of rational numbers might have a least upper bound that is irrational, so that the set doesn’t have a least upper bound in the rational numbers.

sign on its side, and there's a question mark as the isomorphism is the thing we're wondering about.)



The dotted box contains the product we are trying to understand, and we can speculate that perhaps the underlying set of the product is $A \times B$, the product of the underlying sets. So we dream up a “sensible” ordering on $A \times B$ and see if we can exhibit a universal property.

We sort of try this method “optimistically” and find that it works in some individual cases. The idea of category theory is then to make the method into a precise theory and prove it abstractly so that we can use it in more complicated cases without having to rely on optimistic trial-and-error.[†]

We could start by trying this for $\mathbb{R}^2$, since although we talked about $\mathbb{R}^2$ as just a product of sets earlier, $\mathbb{R}$ is naturally a poset with the ordering $\leq$. Is there a natural way to order $\mathbb{R}^2$? One way is to do it *lexicographically*, that is, like in a dictionary: you look at the first coordinate first, and look at the second coordinate second. So everything starting with x -coordinate 1 will be less than everything with x -coordinate 2, regardless of the y -coordinates.

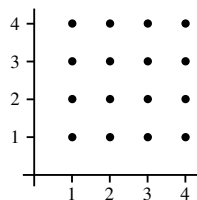
Another way is to consider both coordinates at once and declare this:

$$(x_1, y_1) \leq (x_2, y_2) \iff x_1 \leq x_2 \text{ and } y_1 \leq y_2.$$

In this case *both* coordinates have to be $\leq$ in order for the pair to count as $\leq$.

Things To Think About

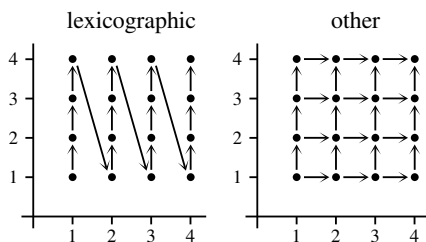
T 18.14 Consider the points in $\mathbb{R}^2$ shown on the right. Try drawing a minimal diagram of arrows between these points, according to the lexicographic ordering and then according to the other ordering we described above. Draw an arrow $a \rightarrow b$ whenever $a \leq b$, but as usual there is no need to draw identities or composites.



Which of your two pictures do you think looks more like a product? Remember, in the definition of a product $A \times B$, A and B play the same role, so in the product picture there should be some sort of symmetry.

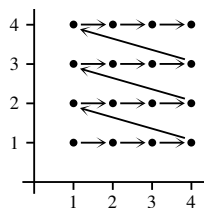
[†] The abstract explanation for this approach involves limits in categories of algebras for monads. This is beyond our scope but I mention it in case you want to look it up later.

According to the lexicographic ordering we know that everything in the column $x = 1$ comes before everything in the column $x = 2$ which comes before everything in the column $x = 3$ and so on. Here are the two diagrams.



I hope you agree that there's something more satisfying about the second diagram. That's a rather subjective point though; a non-subjective point is that the second one is symmetrical in the x and y values, where the first one isn't.

That is, if we switch the x - and y -axes in the second diagram it will stay the same, but if we switch them in the first diagram it will become the one on the right, which is in fact called the *colexicographic* ordering.



Of course, neither the feeling of satisfaction nor the thought about symmetry is enough to prove that the “satisfying” ordering makes that thing into a categorical product.

Things To Think About

T 18.15 What do we need to do to show that this makes $\mathbb{R}^2$ into a product of posets, not just a product of sets?

To show that this is a product we need to exhibit the universal property *in the category of posets* not just in **Set**. We can piggy-back on the universal property in **Set**, but we need to show these things in addition:

1. the projections are now order-preserving functions, not just functions, and
2. when we're inducing a unique factorization, if all the functions we start with are order-preserving then the factorization is also order-preserving.

We might as well state this in generality now, not just for $\mathbb{R} \times \mathbb{R}$.

Proposition 18.4 *Let $(A, \leq)$ and $(B, \leq)$ be posets. Then we can define an ordering $\leq$ on $A \times B$ by $(a_1, b_1) \leq (a_2, b_2) \iff a_1 \leq a_2 \text{ and } b_1 \leq b_2$ and this makes $(A \times B, \leq)$ into a categorical product $(A, \leq) \times (B, \leq)$ in **Pst**.*

One might “abuse notation” and simply say that given posets A and B the product $A \times B$ has ordering as above.

Proof First we show that the projections are order-preserving. Recall that the projections are the functions p and q given on the right.

$$\begin{array}{ccc} A & \xleftarrow{p} & A \times B \xrightarrow{q} B \\ a & \longleftarrow & (a, b) \longmapsto b \end{array}$$

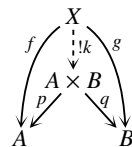
Now to show p and q are order-preserving we need to show

$$(a_1, b_1) \leq (a_2, b_2) \implies \underbrace{p(a_1, b_1) \leq p(a_2, b_2)}_{\text{i.e. } a_1 \leq a_2} \quad \text{and} \quad \underbrace{q(a_1, b_1) \leq q(a_2, b_2)}_{\text{i.e. } b_1 \leq b_2}$$

and this is true by definition of the ordering.

The top parts of the boxes are what we need to show, and the bottom parts together are the definition of the ordering; the top and bottom of each box is equivalent by the definitions of p and q .

Next we need to consider the unique factorization as in the diagram on the right. We know there is a unique factorization k in **Set**; we need to show that if f and g are order-preserving then k is too.



Recall that k is defined by $k(x) = (f(x), g(x))$

so to show that k is order-preserving we need to show

$$x_1 \leq x_2 \implies \underbrace{(f(x_1), g(x_1)) \leq (f(x_2), g(x_2))}_{\text{i.e. } f(x_1) \leq f(x_2) \text{ and } g(x_1) \leq g(x_2)}$$

But this is precisely the definition of f and g being order-preserving. □

Note that this proof doesn't just show that the necessary things are true, it also shows us how the parts of the structure correspond precisely to the things that we need. However, this is a type of proof where I didn't go in with any great "feeling" about how the proof was going to proceed. Rather, I just followed the definitions through, more or less by type-checking. The part that involved "feeling" was in deciding what the appropriate ordering on $A \times B$ would be.

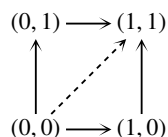
Things To Think About

T 18.16 Do products work like this for totally ordered sets? That is: if we take the product of two tosets regarded as posets, will the result be a toset? Try it for a pair of very small tosets, for example two copies of $0 \rightarrow 1$.

Tosets (totally ordered sets) do not work so neatly. My favorite (or perhaps least favorite) example of this is Tupperware boxes. Rectangular Tupperware boxes have a length and a width, which is like a pair of coordinates (x, y) . We

could order them by either of those things but it's much more useful to be able to stack them, and we can only do that if one of them has *both* its length and its width smaller than the other. Typically this not the case and I end up with a whole load of boxes that won't stack, because the product ordering is only a partial ordering, not a total ordering. At least, that's how I think of it.

For a more formal example: the product of the very small poset $0 \rightarrow 1$ with itself has four elements, ordered as shown (I have drawn in the diagonal to remind us that it is there, but dotted as it's redundant in the diagram).



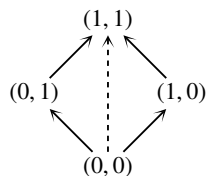
This is not a totally ordered set. It is visibly not in a straight line; more formally we could observe that there is no arrow between $(0, 1)$ and $(1, 0)$.

There are some trivial cases in which a product of tosets will be a toset (if either of the original tosets is empty or has only one element) but in general it will not be a toset. Note that this is an advantage of the lexicographic or colexicographic orderings: we take some tosets and definitely produce a total ordering on the product of the underlying sets. This is useful if we're doing something like making a dictionary and really need a total ordering. However, to get the total ordering we have to sacrifice the universal property. In fact it is quite often the case that a universal structure is impractical in some way and that a non-universal structure will be more useful for practice, where the universal one is more useful for theory.

Note on “in general”

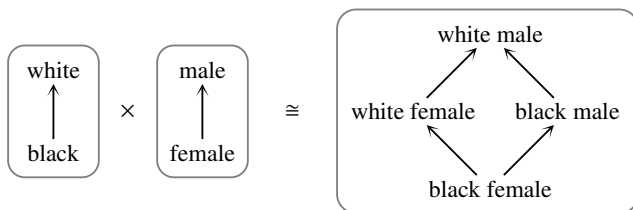
In mathematics, “in general” means something very precise. In normal life it means something more like “most of the time” but in mathematics it means that this is what happens in the absence of special properties. So for tosets, the product is sometimes a toset but only for some special cases of toset. To take an extreme example we might say “ $\frac{1}{x}$ is not defined for numbers in general, but only when $x \neq 0$ ”. In this case it is defined for almost all numbers, but that still doesn't count as “in general”.

The diagram we drew for the product of two copies of the poset $0 \rightarrow 1$ was reminiscent of coordinates, but if we rotate it slightly we can see it as a kind of blueprint for the posets of privilege, specifically the interaction between any two types of privilege.



For example if we consider an individual poset representing the privilege that white people have over black people, and another for the privilege that

male people have over female people, the product poset gives us the square as shown below. It is one face of the cube of privilege.



Things To Think About

T 18.17 What do the projection maps do in this case?

In this case the first projection ignores a person's gender and the second projection ignores a person's race.

There are two forms of antagonism that I see happening in this and analogous situations. One is that those with just one of the two types of privilege tend only to think about the type they lack rather than the type they have. For example, white women only compare themselves with white men and forget how much more privileged they are than black women. The other antagonism is between people who are incomparable in this poset, in this case white women and black men, as they each fight against their own oppression.

This doesn't always happen: some people who lack one type of privilege are then able to empathize with everyone who lacks one (or more) types of privilege, by performing an isomorphism of categories in their head (whether they think of it like that or not).

18.8 Monoids and groups

When we made a product of posets $(A, \leq)$ and $(B, \leq)$ we took the product $A \times B$ of the underlying sets and then developed a partial ordering on it based on the individual orderings. We can do something analogous for monoids and groups, but now we start with groups $(A, \circ)$ and $(B, \circ)$, and instead of making an ordering on $A \times B$ we need to make a binary operation based on the individual binary operations on A and B .

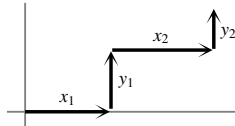
As with posets we can try this first on $\mathbb{R}^2$ as that may be a more familiar context. So far we have treated $\mathbb{R}$ as a set, and then as an ordered set. We can also treat it as a monoid or a group, under addition. The idea for the categorical product $\mathbb{R} \times \mathbb{R}$ is to come up with a definition of addition for ordered pairs, like $(x_1, y_1) + (x_2, y_2)$, based on how we add individual numbers.

Things To Think About

T 18.18 What pair of coordinates could be the answer to $(x_1, y_1) + (x_2, y_2)$? If you're stuck, think about these coordinates not as points, but as an instruction to move a certain direction horizontally and a certain direction vertically.

If we think of (x, y) as telling us to move x units horizontally and y units vertically then it might be more obvious how to add these things together.

The expression $(x_1, y_1) + (x_2, y_2)$ says to move x_1 units horizontally and y_1 vertically, then x_2 horizontally and y_2 vertically, as shown on the right.



The total result is that we have gone $x_1 + x_2$ units horizontally and $y_1 + y_2$ units vertically, so we seem to be saying $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$.

Things To Think About

T 18.19 Check that this binary operation makes $\mathbb{R}^2$ into a group. What is the identity and what are inverses?

We call this doing addition “pointwise” or “componentwise” as we do it on each component separately, just like when we did posets we defined the ordering on the product componentwise. When we do anything pointwise, typically axioms all follow as they hold in each component individually. That is the case here. In particular the identity is $(0, 0)$, and the inverse of (x, y) is $(-x, -y)$.

Now let's generalize this.

Things To Think About

T 18.20 Can you generalize our approach to defining addition on $\mathbb{R}^2$, to define categorical products of monoids in general? For the proof, you could try taking our proof about products of posets as a blueprint.

Proposition 18.5 *Let $(A, \circ, 1)$ and $(B, \circ, 1)$ be monoids. We define a binary operation on $A \times B$ by $(a_1, b_1) \circ (a_2, b_2) = (a_1 \circ a_2, b_1 \circ b_2)$.*

This makes $A \times B$ into a monoid with unit $(1, 1)$, and it is the categorical product of the original two monoids.

The idea of the proof is similar to our proof for posets. We know that we have projections and a universal property in **Set**, and we need to show the following:

1. the projections are now monoid homomorphisms, not just functions, and
2. the factorization induced in **Set** by a pair of monoid homomorphisms is also a monoid homomorphism (see below for further explanation of this).

I strongly believe that after a certain point in this proof it's much easier to write your own proof than try and decipher the symbols involved. I think many (or most or all) mathematicians read research papers this way — reading just enough to try and work out the proof for themselves. But here it is anyway.

Proof First we show that the projections are homomorphisms. As usual the projections are the functions p and q given on the right.

$$\begin{array}{ccc} A & \xleftarrow{p} & A \times B \xrightarrow{q} B \\ a & \longleftarrow & (a, b) \longrightarrow b \end{array}$$

We will deal with p ; then q follows similarly. To show that p is a homomorphism we need to show that it respects the binary operation and the unit.

For the binary operation we need to show $p((a_1, b_1) \circ (a_2, b_2)) = p(a_1, b_1) \circ p(a_2, b_2)$.

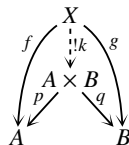
We basically unravel the definitions on both sides and show they're the same. When writing it out formally it might be better to start on the left and work our way to the right, even though in rough we might work on both sides at the same time.

For the binary operation:

$$\begin{aligned} p((a_1, b_1) \circ (a_2, b_2)) &= p(a_1 \circ a_2, b_1 \circ b_2) && \text{by definition of } \circ \text{ in the product} \\ &= a_1 \circ a_2 && \text{by definition of } p \\ &= p(a_1, b_1) \circ p(a_2, b_2) && \text{by definition of } p \end{aligned}$$

For the identity $p(1, 1) = 1$ as needed. So p is a homomorphism, and so is q .

Next we need to consider the unique factorization as in the diagram on the right. We know there is a unique factorization k in **Set**; we need to show that if f and g are homomorphisms then k is too.



Again, we basically just unravel both sides of the definition of homomorphism and find that they're the same.

Recall that k is defined by $k(x) = (f(x), g(x))$. Now:

$$\begin{aligned} k(x_1 \circ x_2) &= (f(x_1 \circ x_2), g(x_1 \circ x_2)) && \text{by definition of } k \\ &= (f(x_1) \circ f(x_2), g(x_1) \circ g(x_2)) && \text{since } f \text{ and } g \text{ respect } \circ \\ &= (f(x_1), g(x_1)) \circ (f(x_2), g(x_2)) && \text{by definition of } \circ \text{ in the product} \\ &= k(x_1) \circ k(x_2) && \text{by definition of } k \end{aligned}$$

so k is a homomorphism. Finally we show that k preserves identities:

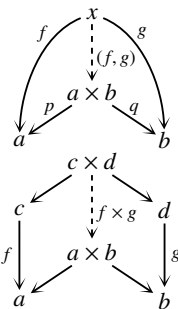
$$\begin{aligned} k(1) &= (f(1), g(1)) && \text{by definition of } k \\ &= (1, 1) && \text{since } f \text{ and } g \text{ each preserve identities} \end{aligned}$$

which is the identity in the product as required. $\square$

The analogous result is true for groups but it's actually slightly easier as we don't have to check for the preservation of identities.

18.9 Some key morphisms induced by products

The standard morphism we've seen induced by a product as shown on the right is often written (f, g) , based on its construction in **Set**.



Another useful morphism induced from a product is shown here. This one is really induced from some composites: we use the composites from $c \times d$ all the way to a and b via f and g respectively to produce a factorization often called $f \times g$.

Things To Think About

T 18.21 See if you can work out what $f \times g$ does in **Set**.

Given functions $C \xrightarrow{f} A$ and $D \xrightarrow{g} B$, the function $f \times g$ has the effect shown on the right.

$$\begin{array}{ccc} C \times D & \longrightarrow & A \times B \\ (c, d) & \longmapsto & (f(c), g(d)) \end{array}$$

I mention this here as we will need maps of this form later, when we talk about products of categories in Section 21.3.

18.10 Dually: coproducts

We have spent quite a long time on products. We are now going to look at the dual concept, which is called a coproduct. Remember “look at the dual” means we look at products in $\mathcal{C}^{\text{op}}$ and then translate the concept back into $\mathcal{C}$.

Definition 18.6 A *coproduct* in a category $\mathcal{C}$ is a product in $\mathcal{C}^{\text{op}}$.

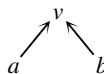
Things To Think About

T 18.22 Can you “translate” this definition back to a concept directly in $\mathcal{C}$?

Unraveled definition of coproduct

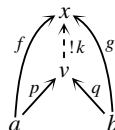
We just need to turn around every arrow in the definition.

Let a and b be objects in a category $\mathcal{C}$. A *coproduct* of a and b is a universal diagram of the shape shown on the right.



Eventually the aim is to be comfortable enough with the general idea of universal properties that this succinct wording is enough, but I'll spell it out here as we do need to be careful about some directions.

The universal property says: given any diagram of the same shape, as in the curved arrows on the right, there is a unique factorization k as shown.



Note that the factorization k now goes *from* the universal vertex v to the vertex x that is not universal, where for products it was the other way round. The key here is not to think of that direction exactly, but to think of the fact that we are “factorizing” the non-universal diagram: expressing it as a composite of the universal one and k . Then some type-checking will ensure that the unique factorization map is pointing in the appropriate direction.

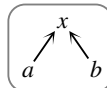
We have some terminology analogous to the product situation:

- p and q are called *coprojections* or *insertions*.
- Instead of a cone, this is now a *cocone*. In a cone we have projections pointing away from the vertex, and in a cocone we have coprojections pointing to the vertex. The coproduct is then the *universal cocone*.
- The coproduct is written $a \coprod b$, $a \sqcup b$, or $a + b$ for reasons we'll see.

Things To Think About

T 18.23 A coproduct is an initial object *somewhere*; can you work out where? It's another kind of “souped-up” category.

A coproduct for a and b is an initial object in a category whose objects are cocones over a and b , that is, diagrams of the form shown on the right. Morphisms are morphisms between the vertices (the top objects) making everything commute.



Things To Think About

T 18.24 What results immediately follow by duality with products?

As coproducts in $\mathcal{C}$ are products in $\mathcal{C}^{\text{op}}$ we have dual versions of everything that is true for products. So coproducts are unique up to unique isomorphism (making cocones commute), there is symmetry in the variables, and there are n -fold versions. Now let's see what coproducts are in various example categories.

18.11 Coproducts in **Set**

For a coproduct of sets A and B , we're looking for a way of producing a canonical set with functions from A and B into it. It turns out to be the *disjoint union*. Disjoint unions have a technically slightly obscure looking definition, but the idea is that you stick the two sets together and ignore the fact that some of the objects might have been the same. You can think of it as painting all the elements of A red and all the elements of B blue to force them to be different, and then taking the “normal” (not disjoint) union.

In case this isn't clear, let's consider these sets: $A = \{a, b, c\}$, $B = \{b, c, d\}$.

When we take the normal union we get this: $A \cup B = \{a, b, c, d\}$

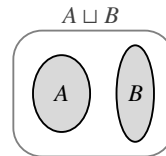
This is because the normal union says that if there are elements in the intersection (that is, in both sets) we don't count them twice.

The *disjoint* union is something like this: $A \sqcup B = \{\text{red: } a, b, c, \text{ blue: } b, c, d\}$

Or perhaps I could use fonts instead of colors: $A \sqcup B = \{a, b, c, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$

For the normal union we can't tell how many elements there will be unless we know how many there are in the intersection; for the disjoint union we just add the number of elements in A and B together (that's if they're finite; if they're infinite then the union or disjoint union will also be infinite).

We can picture the disjoint union as just sticking the sets side by side, as shown here.



The technical definition of disjoint union is: $A \sqcup B = (A \times \{0\}) \cup (B \times \{1\})$.

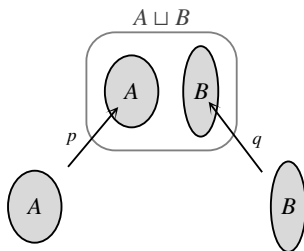
We won't really need it here but it's quite a clever definition and you might like to think about what it's doing.

Things To Think About

T 18.25 Can you show that the disjoint union is a coproduct in **Set**? The first step is just understanding what it is we need to show, and that's a good step even if you can't then show it. You can do it for the specific example above, or for disjoint unions in general; you can use the formal definition if you've understood it, or just get the idea via pictures.

We need to exhibit the coprojections and then verify the universal property.

Here's a picture encapsulating the idea of the coprojections. Note that technically we shouldn't say " $A \sqcup B$ is a coproduct" without saying what the coprojections are, but we often do when the coprojections can be understood to be the "obvious" inclusions.



Personally I find that this picture is much more illuminating than the formal definition of the coprojection functions, but it's important to be able to turn the idea into a rigorous definition if you want to become a rigorous mathematician. However, the idea is also important, and as the rigorous part is much easier to look up elsewhere than the idea, I'll leave it at the picture here.

It remains to check the universal property. Again I think the idea is possibly more important than the formality. The idea is that to define a function *out of* the disjoint union, exactly what you have to do is define it on the A part and also define it on the B part. Those are completely separate issues, so this is the same as defining a function out of A and a function out of B separately. Thus for any set X we have a bijective correspondence between

1. pairs of functions $A \xrightarrow{f} X$ and $B \xrightarrow{g} X$, and
2. functions $A \sqcup B \xrightarrow{k} X$.

Moreover it's not "any old" bijection: the correspondence is produced by composition with the coprojections.

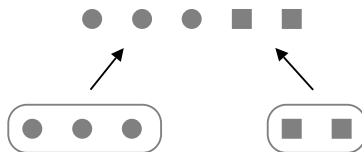
This is an important way of thinking about universal properties. In general it says there is a bijective correspondence between (co)cones and factorizations, so that we can equivalently encapsulate the information of any (co)cone as a single morphism to (or from) a universal one. It's very handy because a cone generally involves many morphisms, and we can now encapsulate it as one morphism together with a "reference" cone.

This is crucially the idea behind the higher level abstract definition of universal property that we can't yet do as we haven't seen natural transformations. It is the root of what category theorists mean when they say that a bijection is "natural" when they use "natural" technically rather than informally.

18.12 Decategorification: relationship with arithmetic

We sometimes use the notation $+$ for coproducts because coproducts of sets correspond to addition of ordinary numbers in a way that we'll now discuss.

When we first introduce addition to small children we might give them a number of objects, and then some more, and see how many there are altogether. So to do $3 + 2$ we give them 3 objects and then 2 more objects and then take the disjoint union of those two sets; we just don't say it like that, typically. The coprojection functions might consist of us physically sliding some small objects into place, as in this picture.



In fact, when we do this physical addition, I think we are doing profound categorical mathematics with children; likewise if we do physical multiplication in grids that we saw earlier on. We are showing them deep structures behind arithmetic, which then get flattened out if we degenerate into drilling “number facts” in the slightly later years in school.

I often say that numbers are an abstraction because we make them by forgetting a lot of details about objects in the real world. However another point of view is that they're a “de-categorification”,[†] regarding isomorphic sets as strictly the same. For example there are infinitely many one-element sets, and they're all isomorphic in the category **Set**. If we regard them as strictly the same there is just one of them and this is, essentially, the number 1. Likewise there are infinitely many two-elements sets and they're all isomorphic, and we make them strictly the same and call that thing the number 2.

Then addition such as $a + b$ is defined by taking any two sets that were turned into the numbers a and b , finding their categorical coproduct, and then turning those into numbers again. This is what counting with little children is like, when you give them objects and ask them to just see how many there are. Multiplication is defined by taking categorical products. This shows that addition and multiplication are dual to each other, which is a curious point of view given that multiplication is also repeated addition.

The characterization as repeated addition ends up being something that can be proved using universal properties together with the fact that every set is a coproduct of one-element sets. Many other “laws” of arithmetic can also be proved by categorifying them, using universal properties in **Set**, and then decategorifying again. This includes commutativity, associativity, and the distributive law of multiplication over addition $a(b + c) = ab + ac$. I think it's a shame that we tend to dismiss “physical arithmetic” as a mere stepping stone to abstract arithmetic.

[†] This was coined in J. Baez and J. Dolan, Categorification. *Higher category theory* (Contemporary Mathematics, No. 230) 1–36, American Mathematical Society, 1998.

18.13 Coproducts in other categories

We will now look at coproducts in the other categories where we've seen products, but only briefly. A general principle is that universal properties involving cocones (rather than cones) are much harder to construct for “sets with algebraic structure”.[†]

Things To Think About

- T 18.26**
1. What are coproducts in a category of factors of n ? Follow through what we did for products but do it dually. Then try and generalize to any category that is a poset.
 2. In the category of posets and order-preserving maps, we found products by taking the product of the underlying sets and then putting an ordering on it. Can we do coproducts like this? Does it work for totally ordered sets?
 3. Coproducts of monoids are hard: disjoint union doesn't work. Why not?

Inside a poset

In the poset of factors of n we saw that products are lowest common multiples. Coproducts are then highest common factors. In more general posets we saw that products are least upper bounds; coproducts are then greatest lower bounds. The fancy word for a least upper bound is *supremum* and the fancy word for a greatest lower bound is *infimum*.

In the category of posets

We can take a disjoint union of two posets A and B and it will still be a poset: everything in the A part will simply be incomparable to everything in the B part. I hope at this point I can leave it to you to check the universal property.

However, as for products if we take the coproduct of two tosets A and B , the result is in general not a toset: the elements of A will be incomparable to the elements of B , and this is not allowed in a toset. As with products it will work in some special cases (if either of the original sets was empty).

In the category of monoids

In the category of monoids if we take the disjoint union of A and B as sets then the result is not in general a monoid — we have a binary operation on the A part and a binary operation on the B part but we do not have a way of doing $a \circ b$ with an element from each side.

[†] In case you're interested in looking it up: this statement can be made precise in terms of algebras for monads.

Sorting this out is somewhat complicated. We don't want the answer to $a \circ b$ to be anything that already exists because that would be non-canonical, so we “throw in” an extra element to be the answer. Essentially this consists of throwing in all ordered pairs $a \circ b$ for any $a \in A$ and $b \in B$.[†] But then what if we try and do the binary operation on one of those new elements together with another element?

For example $(a' \circ (a \circ b))$ would need to be the same as $((a' \circ a) \circ b)$ by associativity, and this is still of the form “element of $A \circ$ element of B ”.

However $((a \circ b) \circ a')$ can't be reduced in general as we don't know that $\circ$ is commutative. Likewise $((a \circ b) \circ (a' \circ b'))$ and so on — in fact we could end up with infinite strings of “stripy” elements, that is, alternating between elements of A and elements of B . These are what we have to construct to make the coproduct of monoids.

If we insist that the binary operation is commutative then things are much simpler. Then in any long string of a 's and b 's the a 's can all commute past the b 's and be gathered together, for example $(a \circ b \circ a' \circ b' = (a \circ a') \circ (b \circ b'))$. So if we restrict to the category of *commutative* monoids everything in a coproduct can indeed be expressed as “element of $A \circ$ element of B ”.

You might notice that this is oddly similar to the elements of the product $A \times B$ apart from in notation, as the elements of the product are

$$(\text{element of } A, \text{element of } B).$$

This is the same information, just with a different symbol in between the elements. This means that in the category of commutative monoids or (more typically) commutative groups, the coproduct and the product are the same. This is unusual, but is a similar phenomenon to the fact that the terminal and initial objects are the same. It is one of the things that makes the category of commutative groups so special. Commutative groups are also called Abelian groups after the mathematician Abel.[‡]

There are some unhelpful issues of terminology around universal properties. One is that coproducts of groups are typically called “free products” in group theory, and products are called “direct products” (there is something else called a “semi-direct product”). It took me ages to work out that the free product of groups was actually the coproduct and I wish someone had told me. Some people think it's pedagogically better if you work these things out for yourself, but I was the kind of student who was convinced that I must be confused and

[†] We're used to writing ordered pairs as (a, b) but writing them $a \circ b$ is the same information just presented differently; we'll come back to this in a moment.

[‡] This gives rise to the ridiculous joke: What's purple and commutes? An Abelian grape.

doing something wrong if I thought of something that hadn't been told to us, so I was afraid to ask, for fear of being thought stupid. One solution to this is to encourage all students to be more arrogant and believe in their own brilliance; another is to explain more things and make sure you never make anyone feel stupid if they ask a question.

18.14 Further topics

Topological spaces

The product of topological spaces follows the same principles as above but is somewhat more technically difficult, and we haven't actually done the technical definition of topological space in the first place. However the idea is the same: we take the cartesian product $A \times B$ of underlying sets, and then try and put a topology on it in a canonical way. As with posets there are several different possibilities, but it's more subtle to see which is the categorical product as it's not just a simple case of symmetry; it's really about universality. The technicalities of what is called the *product topology* are a little subtle, especially if you want the possibility of infinite products, but the ideas are quite vivid in low dimensions at least: the product of two shapes is the result of waving one shape in the shape of the other, and imagining what higher-dimensional space you "sweep out". For example the product of a square and a line is a cube, because if you move a square in a line you sweep out a cube in the air.[†] The product of a circle with a circle is a torus (like the surface of a bagel), which I like to demonstrate by taking a slinky (essentially a circle) and wrapping it round on itself in a circle. You can even make a vertical cut in a bagel and then insert the bagel into the slinky-torus.

Anyway, when we work categorically then we don't really need to know exactly what the formal definition is. We just need to know that the categorical product exists, and then we work with its universal property rather than its definition.

In the category of categories

We can take products and coproducts of categories, and perhaps you can at this point guess what they are. We'll come back to it after we've spent more time with functors and the category **Cat** of (small) categories, where these products and coproducts live.

[†] This does depend on some details such as the line being the same length as the edges of the square, and the "sweeping" motion being exactly at right-angles to the face of the square.

Further analogies with numbers

There are more things we can do inspired by the analogy with products of numbers. We have already taken a passing look at commutativity (the symmetry between a and b in the categorical product $a \times b$) and associativity. Other things we could ask include:

- Is there an identity? (Often yes. What might it be?)
- Are there inverses? (Often no. Why?)
- Do we have something like unique factorization into primes? If so what are the “primes”?

The last one is quite well studied for groups and is particularly straightforward for Abelian groups, where there is a “fundamental theorem of Abelian groups” rather like the fundamental theorem of arithmetic (which says that every natural number can be uniquely expressed as a product of primes).

For the first question we can show that terminal objects are a sort of weak unit for products, in that $a \times 1$ is canonically isomorphic to a for any object a (and dually for coproducts and initial objects). This is then a prototype example of a “monoidal category” which is the categorification of a monoid, that is, it is a monoid but up one dimension. A monoid is a set with a binary operation; a monoidal category is essentially a category with a binary operation. However, in order to make the notion appropriate for its dimension, associativity and unitality on objects does not hold strictly as this would involve equalities between objects. Instead there are canonical isomorphisms similar to the ones we’ve seen for products, but as they are not necessarily induced by a universal property we impose some axioms on them to make sure they behave almost as well as the ones induced by a universal property.

We will touch on this in the last chapter, on higher dimensions.